Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Efficient Agentic AI Systems: LLM-Agent Orchestration and Hardware-Aware Acceleration for Autonomous Enterprise and Edge Automation**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Embedded Software*	*Semiconductor Devices and Systems*	*SoC*
Concept of Proposed Technology	Agentic AI can plan, act, and self-improve by coordinating large language models (LLMs) with tools (APIs, databases, simulators, and code execution), enabling the next wave of AI automation. However, today's LLM-agent deployments are often too expensive, slow, and brittle for SMEs—especially under privacy and on-premises constraints. This proposal will develop an efficiency-first agentic AI stack that reduces total cost, latency, and energy per completed task while improving reliability, safety, and deployability. Core technical components are as follows: • Resource- and budget-aware agent orchestration: planner-executor loops, hierarchical/multi-agent coordination, and dynamic stopping criteria with explicit budgets			

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	(tokens, latency, energy). • Adaptive model routing: small/large model cascades, mixture-of-experts style routing, early-exit, and confidence-based escalation to minimize expensive LLM calls. • Token-efficient long-context and memory: retrieval-augmented generation (RAG), episodic memory, summarization/compression, and bounded-context policies for predictable cost. • Reliability via tool-grounded execution: structured tool-calling, self-checking/verification loops, and deterministic validators to reduce hallucinations and execution errors. • Hardware-aware acceleration for private-edge and Korean NPUs: quantization, KV-cache optimization, batching, and compiler-friendly optimizations; plus ShiftAddLLM (multiplication-less reparameterization), AmoebaLLM (shape-adaptive deployment), and hardware-aware NAS/compression tuned to target NPU instruction sets and memory hierarchies. **Representative SME pilots (co-designed with partner):** 1) Enterprise/on-prem: document processing + knowledge-base Q&A + ticket triage with auditable tool use. 2) Industrial/edge: operations planning or inspection workflows with private-edge deployment and latency/energy budgets. **Korean-market readiness:** Evaluate and optimize on Korean-language tasks and SME data (where permitted), and support integration with Korean LLM endpoints or on-prem models.
Rationale for Joint R&D Project	Joint R&D is essential to translate efficient agentic AI into real products and deployments. **Why collaboration is needed:** • Korean SMEs provide real workflows, toolchains (enterprise apps/APIs/datastores), datasets/logs, privacy & compliance requirements, and deployment targets (cloud, on-prem, edge). • Georgia Tech (EIC Lab) provides world-leading expertise in cross-layer efficiency methods spanning algorithms-systems-hardware (efficient LLM inference/deployment and hardware-aware optimization), backed by strong publication impact and spotlight/best paper awards (e.g., NeurIPS 2025, ECCV 2024, and MICRO 2024). • For private-edge/NPU deployment, effective algorithm-hardware co-design is necessary to achieve competitive inference performance without relying on the NVIDIA CUDA ecosystem. • Iterative co-design and validation on partner environments is required to meet practical constraints (cost, latency, reliability, governance) and to ensure a commercialization-ready prototype. **Expected benefits to Korean SMEs:** • 30-60% reduction in total token usage and compute cost per successful task for multi-step workflows. • >=5x faster end-to-end task completion at the same success criteria via routing/caching/optimization.

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	• >=10-20% absolute improvement in task success rate (or critical error reduction) via verification and tool-grounded execution. • For NPU/private-edge deployments: target >=3x tokens-per-watt and >=5x tokens-per-second for LLM-agent inference with >99% baseline accuracy on partner hardware. • A deployable, privacy-preserving agentic AI product option (SaaS, on-prem, or private-edge) that differentiates SME offerings. **Target Korean SME partner archetypes (examples):** • Enterprise software / AI automation (on-prem or SaaS) integrating LLM agents with CRM/ERP/helpdesk/document systems. • Smart factory / industrial operations SMEs integrating agents with MES, scheduling, inspection, and planning tools. • Edge AI device / NPU SMEs seeking private-edge agent deployments with hardware-aware acceleration. • AI driven smart home systems or robotic systems.
Key Words	1. Efficient LLM-based agentic systems (planning + tool use) 2. Resource-aware orchestration and model routing 3. Token/energy/cost-efficient long-context and memory (RAG, summarization) 4. Hardware-aware acceleration for private-edge AI agents 5. Safe and governable tool-calling

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**R&D Objective:** Develop a production-oriented, efficiency-first agentic AI framework that enables reliable multi-step task automation with substantially reduced cost, latency, and energy, while supporting safe tool use and privacy-preserving deployment. **Specific objectives:** • Build a modular agent runtime (planner-executor + tool calling) with instrumentation for cost/latency/energy and reliability measurement. • Develop efficiency innovations (adaptive routing, caching, early-exit, budget-aware planning, long-context compression) to minimize LLM compute per successful task. • Deliver a Korean NPU/private-edge acceleration module: ShiftAddLLM/AmoebaLLM-based model optimization plus quantization, KV-cache optimization, batching, and compiler/toolchain integration on partner platforms. • Validate on SME-driven workflows and deliver a pilot-ready prototype system (TRL 6) plus an SDK and commercialization/transfer package.			
R&D Target Specifications	Success will be measured on SME-defined benchmark tasks and deployment settings: • **Workflow Cost / Efficiency:** >=40% reduction (target range 30-60%) in total token usage and compute cost per completed task vs. baseline agent pipelines. • **Latency:**>=5x faster end-to-end task completion for representative multi-step workflows under the same success criteria. • **Reliability:** >=15% absolute improvement (target range 10-20%) in task success rate or equivalent reduction in critical errors via verification and tool-grounded execution. • **Deployability**: Support cloud and on-prem/private-edge deployments with configurable privacy, safety, and monitoring policies. • **NPU/private-edge acceleration targets (on partner hardware):** >=3x tokens-per-watt and >=5x tokens-per-second vs. standard FP16 baseline while maintaining >99% baseline accuracy.			
R&D Plan	This 3-year project follows a build-measure-deploy strategy with continuous SME feedback. **Year 1: SME Use-Case Definition, Baselines, Core Agent Runtime, and NPU Characterization** • Select 1-2 SME-driven pilot workflows (enterprise and/or edge) and define success metrics. • Build a benchmark suite and instrumentation for cost/latency/energy and reliability measurement. • Implement core orchestration: planner-executor, tool-calling protocols, retrieval/memory, and verification loops. • Establish baselines (single vs multi-agent, RAG vs non-RAG, small vs large models) and identify bottlenecks. • Characterize partner Korean NPU architectures (instruction sets, memory hierarchy) and build profiling harnesses.			

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	Year 2: Efficiency + Safety Innovations, Toolchain Integration, and Pilot Evaluations • Develop adaptive routing, caching, early-exit, budget-aware planning, and long-context compression. • Implement governance layer: policy constraints, sandboxed tool execution, audit logs, monitoring dashboards. • Integrate ShiftAddLLM/AmoebaLLM and hardware-aware compression/quantization into partner compiler toolchains and runtimes. • Optimize inference for private-edge deployments (KV-cache optimization, batching, memory access). • Integrate with SME toolchains (APIs, databases, enterprise apps) and run pilot evaluations in staging environments. **Year 3: TRL 6 Pilot Prototype, SDK Release, and Technology Transfer** • Deliver a packaged agentic system with deployment playbooks (SaaS/on-prem/private-edge) and reliability monitoring. • Conduct field trials in SME environments; refine robustness, UX, and compliance requirements. • Release a production-ready SDK (agent stack + NPU optimization toolkit) with documentation and reference configurations. • Technology transfer: licensing-ready code, partner enablement, and commercialization roadmap.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the development of core AI algorithms and system-level intelligence for the proposed efficient agentic AI system. In particular, Georgia Tech (EIC Lab) will be responsible for: • Designing the agentic framework and efficiency algorithms (routing, budgeting, memory, verification). • Building benchmarking methodology and instrumentation for cost/latency/energy and reliability measurement. • Implementing governance/safety layers (policy engine, sandboxing, audit logs) and reference deployments. • Developing and integrating hardware-aware optimization for Korean NPUs, including ShiftAddLLM/AmoebaLLM, compression/quantization, KV-cache optimization, batching, and compiler/toolchain integration. • Supporting IP generation and knowledge transfer to SME engineering teams. Technical guidance and collaboration with the industrial partner to support system integration and experimental evaluation.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for: • Providing target workflows, tool access (APIs/databases/enterprise apps), and success criteria for real deployments. • Supplying representative datasets/logs and constraints (privacy, compliance, latency/energy budgets). • Providing deployment targets (cloud, on-prem, or edge hardware) and supporting integration/testing. • For NPU/private-edge partners: providing NPU hardware development kits, architecture specifications, and compiler toolchain/runtime access; running on-hardware profiling and benchmarking. • Integrating the agentic system into products and conducting user/field evaluations.

	• Defining commercialization requirements (integration, UX, reliability targets, go-to-market feedback).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver efficient agentic AI systems that can plan, act, and self-improve by coordinating large language models (LLMs) with tools (APIs, databases, simulators, code execution) on the target edge platforms/applications. The project will advance the state of the art in efficient but capable agentic AI systems that can benefit numerous edge applications. Expected outcomes will include: • A modular agentic AI software stack for efficient multi-step task automation, suitable for commercialization (SaaS, on-prem, or private-edge). • A benchmark suite and instrumentation tools for measuring task-level cost, latency, energy, reliability, and policy compliance on SME workflows. • A Korean NPU/private-edge acceleration toolkit (SDK + reference configurations) enabling efficient LLM-agent inference via model optimization and compiler/runtime integration. • Pilot deployments demonstrating substantial reductions in operational cost and faster task completion with improved success rates. • A safety/governance layer enabling controllable tool use with auditability, supporting privacy-preserving deployments. • Potential joint IP on resource-aware orchestration, model routing, verification, long-context efficiency, and hardware-aware deployment optimizations. **Broader impact**: enabling Korean SMEs to deploy economically viable, privacy-preserving 'green' AI agents; and strengthening Korean NPU/edge AI ecosystems with competitive software tooling. From an industrial perspective, this project will provide the Korean SME partner with core techniques that enable next-generation efficient agentic AI systems. The outcomes will support **technology commercialization and product development**, enabling cost-effective and scalable agentic AI solutions for domestic applications. More broadly, the project is expected to strengthen the competitiveness of Korean AI technology SMEs and contribute to the growth of AI-driven innovative edge applications.

6. References

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☐ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization